datascope

Data Quality Diagnostic Report

Source: sample_sales.xlsx

August 10, 2026

v2.4.0

0	2	3
Critical	**Warning**	**Info**

5 findings across 15 columns and 500 rows

Executive Summary

No critical issues were found, but 2 warnings and 3 informational observations were detected. Warnings at this level typically skew joins and aggregations quietly — address them before this data feeds reporting.

Detailed Findings

Warning (2)

Warning	mostly_null	Missing Values
Assumption	Column 'mostly_null' is expected to be fully populated.	
Reality	However, 412 of 500 rows (82.4%) are null or blank. The missing values are scattered across the dataset.	
Impact	With more than half its values missing, 'mostly_null' is unreliable for analysis. Aggregations will silently exclude the missing rows, and any model trained on this column will learn from a biased sample.	
Recommended Fix	Investigate why 'mostly_null' has missing values. If the blanks represent a known condition, consider a default value or a separate status column. If they are data-entry gaps, backfill from the source system.	
Prevention Rule	Add a NOT NULL or completeness check for 'mostly_null' at ingestion time. Flag any batch where null rate exceeds the historical baseline.	

Warning	notes	Missing Values
Assumption	Column 'notes' is expected to be fully populated.	
Reality	However, 250 of 500 rows (50.0%) are null or blank. The missing values are scattered across the dataset.	
Impact	With more than half its values missing, 'notes' is unreliable for analysis. Aggregations will silently exclude the missing rows, and any model trained on this column will learn from a biased sample.	
Recommended Fix	Investigate why 'notes' has missing values. If the blanks represent a known condition, consider a default value or a separate status column. If they are data-entry gaps, backfill from the source system.	
Prevention Rule	Add a NOT NULL or completeness check for 'notes' at ingestion time. Flag any batch where null rate exceeds the historical baseline.	

Info (3)

Info	constant_col	Near-Constant Column
Assumption	Column 'constant_col' is expected to carry meaningful, varying data.	
Reality	However, only 1 distinct value was found across 500 rows (uniqueness ratio: 0.2%). Most common: 'FIXED' (500 times).	
Impact	A near-constant column like 'constant_col' adds no analytical value. Including it in models or reports may mislead readers into thinking the field varies when it does not.	
Recommended Fix	Verify whether 'constant_col' should actually vary. If not, document it as a constant and consider removing it from analysis. If it should vary, investigate why the data is uniform.	
Prevention Rule	If 'constant_col' is supposed to carry diverse values, add a data-quality check that flags columns with fewer than 1% unique values.	

Info	discount_pct	Missing Values
Assumption	Column 'discount_pct' is expected to be fully populated.	
Reality	However, 80 of 500 rows (16.0%) are null or blank. The missing values are scattered across the dataset.	
Impact	Missing values in 'discount_pct' will be silently excluded from calculations. If the missingness is not random, aggregations and models will be biased toward the non-missing subset.	
Recommended Fix	Investigate why 'discount_pct' has missing values. If the blanks represent a known condition, consider a default value or a separate status column. If they are data-entry gaps, backfill from the source system.	
Prevention Rule	Add a NOT NULL or completeness check for 'discount_pct' at ingestion time. Flag any batch where null rate exceeds the historical baseline.	

Info	mostly_null	Near-Constant Column
Assumption	Column 'mostly_null' is expected to carry meaningful, varying data.	
Reality	However, only 1 distinct value was found across 88 rows (uniqueness ratio: 1.1%). Most common: 'value' (88 times).	
Impact	A near-constant column like 'mostly_null' adds no analytical value. Including it in models or reports may mislead readers into thinking the field varies when it does not.	
Recommended Fix	Verify whether 'mostly_null' should actually vary. If not, document it as a constant and consider removing it from analysis. If it should vary, investigate why the data is uniform.	
Prevention Rule	If 'mostly_null' is supposed to carry diverse values, add a data-quality check that flags columns with fewer than 1% unique values.	

Field Inventory

Summary of all fields with detected issues.

Field Name	Issue Types	Highest Severity	Finding Count
constant_col	Near-Constant Column	**Info**	1
discount_pct	Missing Values	**Info**	1
mostly_null	Missing Values, Near-Constant Column	**Warning**	2
notes	Missing Values	**Warning**	1